AWS CLOUD WORKSHOP · QUESTION 2

Custom VPC with an Ubuntu Web Server

VPC, public subnet, Internet Gateway, EC2 and Nginx in Mumbai (ap-south-1)

Components

Component	Role
VPC	Isolated private network (10.0.0.0/16)
Public subnet	Address range for the server (10.0.1.0/24)
Internet Gateway	Door between the VPC and the internet
Route table	Sends 0.0.0.0/0 traffic to the gateway
Security group	Firewall allowing SSH (22) and HTTP (80)
EC2 and Nginx	Ubuntu server that serves the HTML page

Network setup

1. VPC

my-web-vpc with CIDR 10.0.0.0/16, created as VPC only.

2. Public subnet

Public-Subnet 10.0.1.0/24 with auto-assign public IPv4 enabled.

3. Internet Gateway

my-web-IGW created and attached to my-web-vpc.

4. Route table

Public-Route-Table with 0.0.0.0/0 to the gateway, associated with the subnet.

Firewall and server

5. Security group web-sg

SSH (22) from My IP and HTTP (80) from 0.0.0.0/0, created in my-web-vpc.

6. EC2 instance

Ubuntu-Web-Server, Ubuntu 24.04 LTS, t3.micro, in Public-Subnet with a public IP, web-sg and a .pem key pair.

Install Nginx

```
ssh -i "web-key.pem" ubuntu@13.206.86.123
sudo apt update
sudo apt install -y nginx
sudo systemctl enable --now nginx
systemctl status nginx
```

Status shows **active (running)**, then the page is written to /var/www/html/index.html.

Result

http://13.206.86.123 shows “Hello from my custom AWS VPC!”

Browser → Internet Gateway → Route table → Public subnet → Security group (port 80) → EC2 → Nginx

Troubleshooting and cleanup

Problem	What to check
SSH times out	SSH source in web-sg, public IP, route to the gateway, network blocking port 22
Page does not load	Port 80 rule, http not https, Nginx running
Permission denied	User ubuntu and the correct .pem file

Cleanup: terminate the instance, then delete the security group, gateway, subnet and VPC to avoid charges.